



UNIVERSITY OF COLOMBO, SRI LANKA



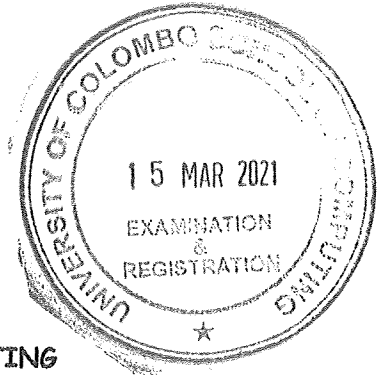
UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination – Semester II – 2020/2021

SCS 1211 – Mathematical Methods I

TWO (2) HOURS



Important Instructions to candidates:

1. The medium of instructions and questions is **English**.
2. **Write your answers to Question No. 1 and Question No. 2 in the BUBBLE SHEET provided. Write your answers to Question No. 3 and Question No. 4 in English in the papers provided.**
3. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
4. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
5. Write your index number on each and every page of the answer script.
6. This paper has 4 questions and 8 pages.
7. Answer **ALL** questions. All questions carry equal marks (25 marks).
8. **This paper consists of two parts, Part A (Question No. 1 and Question No. 2) and Part B (Question No. 3 and Question No. 4). Submit your answers to Part A and Part B together. Attach the bubble sheet to the end of the answer sheets to Part B.**
9. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
10. **Non-Programmable** calculators are allowed.

PART A

Answer all questions. Fill the circle corresponds to the correct response with a pencil on the bubble sheet.

1. (i) There do not exist 4-square matrices A, B such that $|A| + |B| = 10$ and
 - (a) $|AB| = 26$.
 - (b) $|AB| = 24$.
 - (c) $|AB| = 21$.
 - (d) $|AB| = 16$.
- (ii) There is a nonzero $m \times n$ matrix, where m, n are positive integers, such that
 - (a) $\text{trace}(A^T A) = -mn$.
 - (b) $\text{trace}(A^T A) = -(m + n)$.
 - (c) $\text{trace}(A^T A) = 0$.
 - (d) $\text{trace}(A^T A) = 12$.
- (iii) The correct statement is
 - (a) There is a homogeneous system of m equations in n unknowns such that the system is inconsistent.
 - (b) Every homogeneous system of m equations in n unknowns is consistent.
 - (c) There is a homogeneous system of m equations in n unknowns such that the trivial solution is not a solution of it.
 - (d) Every homogeneous system of m equations in n unknowns has a unique solution.
- (iv) Let A, B be 3×3 matrices with integer entries such that A is nonsingular and $AB = BA = |A|I_3$. Then it is possible that
 - (a) $|B| = -5$.
 - (b) $|B| = 6$.
 - (c) $|B| = 0$.
 - (d) $|B| = 16$.

(v) Let $A = \begin{bmatrix} -1 & 5 & 3 & -2 & 4 & 7 \\ 0 & 0 & 1 & 4 & 0 & 0 \\ -7 & 0 & 8 & -6 & 0 & 4 \\ 1 & 0 & 3 & 2 & 0 & 0 \\ -3 & 3 & 5 & -4 & -2 & 8 \\ 0 & 0 & 2 & 5 & 0 & 0 \end{bmatrix}$.

Then $|A|$ is

- (a) -104 .
 - (b) 24 .
 - (c) -264 .
 - (d) 0 .
- (vi) Let A, X be 3×3 matrices with integer entries. Then $\text{trace}(AX + XA) = 7$ has
- (a) a unique solution for X .
 - (b) 3 solutions for X .
 - (c) no solution for X .
 - (d) infinitely many solutions for X .

- (vii) Consider the system of equations with a real constant α .

$$x_1 - 4x_2 + x_3 = -4$$

$$2x_1 + \alpha x_2 - x_3 = 5$$

$$\alpha x_1 + 2x_2 - 5x_3 = -8$$

Then for each real number α , the system

- (a) has a unique solution.
- (b) does not have a unique solution.
- (c) has infinitely many solutions.
- (d) is inconsistent.

(viii) Let A be a n -square matrix. Then

- (a) there is one and only one way to write A as the sum of a symmetric matrix and a skew symmetric matrix.
- (b) there are an infinite number of ways to write A as the sum of a symmetric matrix and a skew symmetric matrix.
- (c) the number of ways A can be written as the sum of a symmetric matrix and a skew symmetric matrix is greater than one and that depends on A .
- (d) A can be written as the sum of a symmetric matrix and a skew symmetric matrix if and only if $A = O_n$.

(ix) Let $A = \begin{bmatrix} 1 & -2 & 0 & -5 & 3 \\ 2 & 0 & 4 & 2 & 2 \\ 3 & 1 & 7 & 6 & 2 \\ -1 & 2 & 0 & 5 & -3 \end{bmatrix}$.

A basis for the column space of A is

(a) $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\}$.

(b) $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 2 \\ -3 \end{bmatrix} \right\}$.

(c) $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ 2 \\ 6 \\ 5 \end{bmatrix} \right\}$.

(d) $\left\{ \begin{bmatrix} 0 \\ 4 \\ 7 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ 2 \\ 6 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 2 \\ -3 \end{bmatrix} \right\}$.

(x) A basis for the null space of $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 0 & 3 & 1 \end{bmatrix}$ is

(a) $\left\{ \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix} \right\}.$

(b) $\left\{ \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} \right\}.$

(c) $\left\{ \begin{bmatrix} 1 \\ -1 \\ -3 \end{bmatrix} \right\}.$

(d) $\left\{ \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix} \right\}.$

2. (i) Let $A = \begin{bmatrix} 1 & 5 & 4 & 7 \\ 3 & -1 & -4 & 5 \\ -2 & 0 & 2 & -4 \\ 0 & 2 & 2 & 2 \\ 4 & -3 & -7 & 5 \end{bmatrix}.$

A basis for the row space of A is

(a) $\{[1, 5, 4, 7], [3, -1, -4, 5]\}.$

(b) $\{[1, 5, 4, 7], [3, -1, -4, 5], [-2, 0, 2, -4]\}.$

(c) $\{[1, 5, 4, 7], [3, -1, -4, 5], [0, 2, 2, 2]\}.$

(d) $\{[1, 5, 4, 7], [3, -1, -4, 5], [4, -3, -7, 5]\}.$

(ii) Let $A = \begin{bmatrix} 5 & -12 \\ 2 & -5 \end{bmatrix}$. Then A^{101} is equal to

(a) $\begin{bmatrix} 5 & -12 \\ 2 & -5 \end{bmatrix}$.

(b) $\begin{bmatrix} -5 & -12 \\ 2 & 5 \end{bmatrix}$.

(c) $\begin{bmatrix} 5 & 12 \\ -2 & -5 \end{bmatrix}$.

(d) $\begin{bmatrix} -5 & 12 \\ -2 & 5 \end{bmatrix}$.

(iii) Let $A = \begin{bmatrix} -16 & 20 \\ -25 & 29 \end{bmatrix}$. Then

(a) $(A^{1/2})^2 = A$ and $(A^2)^{1/2} = A$.

(b) $(A^{1/2})^2 = A$ and $(A^2)^{1/2} \neq A$.

(c) $(A^{1/2})^2 \neq A$ and $(A^2)^{1/2} = A$.

(d) $(A^{1/2})^2 \neq A$ and $(A^2)^{1/2} \neq A$.

(iv) Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{bmatrix}$. Then A has

(a) an eigenspace of dimension 3.

(b) an eigenspace of dimension 2.

(c) an eigenspace of dimension 1.

(d) two eigenspaces of dimension 1 and dimension 2.

- (v) Let V, W be vector spaces over $\mathbb{R}$ of dimensions n, m respectively. Also let T be a linear transformation from V into W . Then there is a matrix A such that $T(x) = Ax$ for each $x \in V$ and A is of order
- (a) $m \times n$.
 - (b) $n \times m$.
 - (c) $n \times n$.
 - (d) $m \times m$.
- (vi) Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over $\mathbb{R}$. Let S, T be subspaces of V such that $S \perp T$ (i.e. for each $s \in S$, for each $t \in T$, $\langle s, t \rangle = 0$). Then
- (a) $S \cup T = V$.
 - (b) $S \cap T = \{0\}$.
 - (c) $S \cup T$ is a subspace of V .
 - (d) $S \cap T \neq \{0\}$.
- (vii) Consider the 4 points; $P_1 \equiv (1, -1, 2)$, $P_2 \equiv (3, 0, 3)$, $P_3 \equiv (0, 1, 5)$, $P_4 \equiv (2, 2, 6)$ in $\mathbb{R}^3$. Then
- (a) the volume of the parallelepiped with vertices P_1, P_2, P_3, P_4 is 9.
 - (b) the volume of the parallelepiped with vertices P_1, P_2, P_3, P_4 is 4.
 - (c) the points P_1, P_2, P_3, P_4 are coplanar but are not collinear.
 - (d) the points P_1, P_2, P_3, P_4 are collinear.
- (viii) Let $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$. Then
- (a) $\text{rank} A = 3$ and $\text{nullity} A = 1$.
 - (b) $\text{rank} A = 2$ and $\text{nullity} A = 2$.
 - (c) $\text{rank} A = 1$ and $\text{nullity} A = 3$.
 - (d) $\text{rank} A = 4$ and $\text{nullity} A = 0$.

(ix) Let $A = \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 3 \\ -4 & 2 \end{bmatrix}$. Suppose that X is a matrix such that

$$AX = B. \text{ Then } X =$$

(a) $\begin{bmatrix} 7/5 & 2/5 \\ 4/5 & 9/5 \end{bmatrix}.$

(b) $\begin{bmatrix} 7/5 & 2/5 \\ -4/5 & 9/5 \end{bmatrix}.$

(c) $\begin{bmatrix} 7/5 & -2/5 \\ 4/5 & 9/5 \end{bmatrix}.$

(d) $\begin{bmatrix} 2/5 & 7/5 \\ 9/5 & 4/5 \end{bmatrix}.$

(x) Consider the system of homogeneous equations

$$2x_1 + x_2 - x_3 + 3x_4 = 0$$

$$2x_1 + x_2 + x_3 + 4x_4 = 0$$

$$6x_1 + 3x_2 - x_3 + 10x_4 = 0$$

$$4x_1 + 2x_2 + 0x_3 + 7x_4 = 0.$$

The incorrect statement is

(a) $x_1 = -s + 7t, x_2 = 2s, x_3 = 2t, x_4 = -4t$ is a solution of the system for each $s, t \in \mathbb{R}$.

(b) $x_1 = -\frac{s}{2} + 7t, x_2 = s, x_3 = 2t, x_4 = -4t$ is a solution of the system for each $s, t \in \mathbb{R}$.

(c) $x_1 = -\frac{s}{2} + \frac{7}{2}t, x_2 = s, x_3 = t, x_4 = -2t$ is a solution of the system for each $s, t \in \mathbb{R}$.

(d) $x_1 = -s + \frac{7}{2}t, x_2 = s, x_3 = t, x_4 = -4t$ is a solution of the system for each $s, t \in \mathbb{R}$.

PART B

3. (i) Let A be a nonsingular n -square matrix. Prove that $\text{adj} A$ is nonsingular and $|\text{adj}[\text{adj} A]| = |A|^{(n-1)^2}$.

- (ii) Let $A = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Express A in the form PDP^{-1} , where D is a diagonal matrix and P is a nonsingular matrix. Is $A^{2021} = A$? Prove your answer. Compute A^{2020} .

- (iii) Let V be the vector space of $n \times n$ matrices under matrix addition and scalar multiplication.

Define $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ by $\langle A, B \rangle = \text{trace}(AB^T)$ for each $A, B \in V$.

Prove that $\langle \cdot, \cdot \rangle$ is an inner product on V .

4. (i) Let V, W be vector spaces over $\mathbb{R}$ and let T be a linear transformation from V into W . Show that $\text{Range}(T)$ is a subspace of W .

- (ii) Let $S = \{e_1, e_2, e_3\}$ be the standard basis for $\mathbb{R}^3$, where $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$ and $e_3 = (0, 0, 1)$. Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $T(x, y, z) = (x - y, y - x, x - z)$ for each $(x, y, z) \in \mathbb{R}^3$, with respect to the standard basis. Compute $[T]_{SS}$. Find a basis for $\text{Ker}(T)$ and a basis for $\text{Range}(T)$.

- (iii) Let $\mathcal{B} = \{u_1, u_2, u_3\}$ be the basis for $\mathbb{R}^3$, where $u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$,

$u_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$. Find $[T]_{\mathcal{B}\mathcal{B}}$, where T is the linear transformation given above in

(ii).